

Qualification Pack



Sewage Treatment Plant (STP) Operator

QP Code: PSC/Q0601

Version: 1.0

NSQF Level: 4

Water Management and Plumbing Skill Council || Unit No-606, DLF Prime Tower
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PSC/Q0601: Sewage Treatment Plant (STP) Operator

Brief Job Description

The Sewage Treatment Plant Operator is responsible for the operation, monitoring, and maintenance of sewage treatment plants. They ensure that wastewater is treated effectively and in compliance with environmental regulations. Sewage Treatment Plant Operators perform various tasks to control and maintain the treatment process, troubleshoot issues, and ensure the plant's efficient and safe operation.

Personal Attributes

The individual must be meticulous and have a keen eye for detail. He / She must have problem solving skills, technical aptitude, adequate knowledge of various treatment methods, equipment operation, and the ability to interpret and analyze data, adaptable and flexible to adjust the approach, prioritize tasks, and respond effectively to unexpected circumstances.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSC/N0604: Operating Sewage Treatment Plant \(STP\)](#)
2. [PSC/N0601: Monitoring and Managing Sewage Treatment Plant](#)
3. [PSC/N0606: Solid Waste Management](#)
4. [PSC/N0603: Maintain Records & Reporting at Site](#)
5. [PSC/N0602: Health and Safety Practices at the treatment Plant](#)
6. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
Country	India
NSQF Level	4

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Credits	15
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3132.0400
Minimum Educational Qualification & Experience	12th Class OR Previous relevant Qualification of NSQF Level (Qualification of NSQF Level 3.5) with 1.5 years of experience relevant experience in Water Supply/ Water Treatment/ Quality Control OR Previous relevant Qualification of NSQF Level (Qualification of NSQF Level 3.0) with 3 Years of experience relevant experience in Water Supply/ Water Treatment/ Quality Control OR 11th Class with 1.5 years of experience Relevant Experience in Water Supply/water treatment/quality control
Minimum Level of Education for Training in School	12th Class
Pre-Requisite License or Training	NA
Minimum Job Entry Age	20 Years
Last Reviewed On	NA
Next Review Date	27/08/2027
NSQF Approval Date	27/08/2024
Version	1.0
Reference code on NQR	QG-04-TH-02938-2024-V1-WMPS
NQR Version	1.0

Remarks:

NA

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PSC/N0604: Operating Sewage Treatment Plant (STP)

Description

this unit is about operating the sewage treatment plant and understanding the various types of unit operations in sewage treatment plant. this unit is also about different terminologies that are used in stp. this unit also covers actively managing the treatment processes which includes adjusting process parameters, controlling equipment operation, and monitoring treatment efficiency

Scope

The scope covers the following :

- prepare for sewage treatment.
- carry out operations in stp

Elements and Performance Criteria

prepare for sewage treatment

To be competent, the user/individual on the job must be able to:

- PC1.** able to start the plant by switching on pump, mixer, controllers and other equipment.
- PC2.** perform the manual work in sewage treatment plant
- PC3.** able to check, open or close the valves
- PC4.** perform the operations at outflow & inflow of the plant by controlling the discharge.
- PC5.** able to collect the samples and facilitate the calibrations of samples.

Carry out operations in STP

To be competent, the user/individual on the job must be able to:

- PC6.** monitor and perform various operations in sewage treatment plant.
- PC7.** perform the mixing of chemicals if required.
- PC8.** handle the sludge properly that is being accumulated properly.
- PC9.** perform the operations involved in purification of gas in scrubber unit evolved from sludge digesters.
- PC10.** complete the documentation work as required.
- PC11.** able to check and clean the bar screen at frequent intervals
- PC12.** able to ensure no large gaps are formed due to corrosion of the screen
- PC13.** able to ensure no large gaps are formed due to corrosion of the screen
- PC14.** able to ensure that the air flow/ mixing is uniform over the entire floor of the tank.
- PC15.** able to adjust the placement of diffusers and the air-flow rate as needed.
- PC16.** able to manually evacuate settled muck/ sediments at least once in a year
- PC17.** able to check oil in the pump every day; top up if necessary
- PC18.** • able to check motor-to-pump alignment after every
• dismantling operation
- PC19.** identify the condition of coupling and replace damaged parts immediately

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- PC20.** identify the vibrations and tighten the anchor bolts and other fasteners
- PC21.** check condition of bearings, oil seals, mechanical seal and replace if necessary
- PC22.** able to follow the manufacturer's O&M manual with diligence.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** • importance and need of treating the water.
•
- KU2.** basic terminologies like discharge, flow velocity, settling velocity, turbulence, plan area organic loading rate, hydraulic loading rate, over flow rate, recirculation, efficiency.
- KU3.** physical characteristics of water like total solids, suspended solids, settleable solids, etc
- KU4.** chemical characteristics of water like ph chloride content, nitrogen content, fats, oil & grease, sulphates, hydrogen sulphide, dissolved oxygen, etc
- KU5.** biological characteristics of water like cod, bod, nitrification, de-nitrification, etc
- KU6.** decomposition by aerobic, anaerobic & facultative bacteria, neutralization, absorption, adsorption, oxidation/reduction, ion exchange process.
- KU7.** disposal of sewage effluents.
- KU8.** different operations performed in treatment of sewage like screening, grit chamber, flow equalization chamber, detritus tank, aeration tank, skimming tank, primary sedimentation tank, coagulation aided sedimentation
- KU9.** Biological operations like sludge digestion, trickling filters, rotatory biological contractors, membrane bio reactors, moving bed bio reactors, septic tank, activated sludge process, aber, waste stabilization pond, planted and unplanted drying beds, secondary sedimentation tank, sludge drying beds, 3rd degree treatment.
- KU10.** process of purifying gas in scrubber unit like purifying gas evolved during sludge digestion process.
- KU11.** method to clean different units present in sewage treatment plant.
- KU12.** handling of chemicals.
- KU13.** process of sewage treatment performed during treatment of sewage.
- KU14.** different methods of handling the sludge.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or hindi/english
- GS2.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** ability to read from different sources- books screens in machines and signage.
- GS4.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS5.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or hindi

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- GS6.** state information, doubts and concerns about work related matters in local language or hindi/english
- GS7.** plan one's daily tasks to achieve maximum efficiency
- GS8.** be punctual and work as per agreed priorities
- GS9.** manage distractions and maintain workplace discipline
- GS10.** listen to the higher authorities carefully
- GS11.** be courteous
- GS12.** identify ways to increase efficiency and reduce errors
- GS13.** read and understand manuals, health and safety instructions, memos, other company documents.
- GS14.** communicate with supervisor.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>prepare for sewage treatment</i>	10	20	-	3
PC1. able to start the plant by switching on pump, mixer, controllers and other equipment.	-	-	-	-
PC2. perform the manual work in sewage treatment plant	-	-	-	-
PC3. able to check, open or close the valves	-	-	-	-
PC4. perform the operations at outflow & inflow of the plant by controlling the discharge.	-	-	-	-
PC5. able to collect the samples and facilitate the calibrations of samples.	-	-	-	-
<i>Carry out operations in STP</i>	20	40	-	7
PC6. monitor and perform various operations in sewage treatment plant.	-	-	-	-
PC7. perform the mixing of chemicals if required.	-	-	-	-
PC8. handle the sludge properly that is being accumulated properly.	-	-	-	-
PC9. perform the operations involved in purification of gas in scrubber unit evolved from sludge digesters.	-	-	-	-
PC10. complete the documentation work as required.	-	-	-	-
PC11. able to check and clean the bar screen at frequent intervals	-	-	-	-
PC12. able to ensure no large gaps are formed due to corrosion of the screen	-	-	-	-
PC13. able to ensure no large gaps are formed due to corrosion of the screen	-	-	-	-
PC14. able to ensure that the air flow/ mixing is uniform over the entire floor of the tank.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. able to adjust the placement of diffusers and the air-flow rate as needed.	-	-	-	-
PC16. able to manually evacuate settled muck/sediments at least once in a year	-	-	-	-
PC17. able to check oil in the pump every day; top up if necessary	-	-	-	-
PC18. • able to check motor-to-pump alignment after every • dismantling operation	-	-	-	-
PC19. identify the condition of coupling and replace damaged parts immediately	-	-	-	-
PC20. identify the vibrations and tighten the anchor bolts and other fasteners	-	-	-	-
PC21. check condition of bearings, oil seals, mechanical seal and replace if necessary	-	-	-	-
PC22. able to follow the manufacturer's O&M manual with diligence.	-	-	-	-
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0604
NOS Name	Operating Sewage Treatment Plant (STP)
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	4
Credits	6
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

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PSC/N0601: Monitoring and Managing Sewage Treatment Plant

Description

this unit involves overseeing the entire treatment process. it also includes regularly monitoring process parameters such as flow rates, pH levels, dissolved oxygen, and solids concentration. Sampling and analyzing wastewater to assess treatment efficiency and quality. this unit aims for continuously optimizing plant performance to achieve effective and efficient sewage treatment.

Scope

The scope covers the following :

- monitoring the stp
- managing the stp

Elements and Performance Criteria

monitoring the stp

To be competent, the user/individual on the job must be able to:

- PC1.** able to monitor the ph, coagulation and flocculation, velocity of flow, discharge at inflow & outflow.
- PC2.** monitoring weather the biological process is being carried out as per the standards.
- PC3.** identify the mechanical instruments such as blowers, pumps, centrifugal separators, valves, aerators, etc are working properly
- PC4.** able to monitor the accumulation of sludge and pressure and temperature of digesters during evolution of gases while sludge digestion as per standards.
- PC5.** identify all the units in sewage treatment plant are in working condition.
- PC6.** able to monitor and record all meter and gauge readings.

managing the stp

To be competent, the user/individual on the job must be able to:

- PC7.** able to identify that the screens are not clogged
- PC8.** identify any kind of leakage at joint of the screen and water channel.
- PC9.** able to examine the equalization tank that it should not overflow or leaking, mixing & aeration system.
- PC10.** able to identify the improper working of aeration system, mixing, leakage at any point, etc...,
- PC11.** able to prepare and maintain the record of operation carried out by sewage treatment plant.
- PC12.** able to fill out the daily activity log sheets.
- PC13.** able to monitor the quality of water at the different units of the plants.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** schematic diagram and working principle of sewage treatment plant.
- KU2.** different types of mechanical instruments such as pumps, blowers, centrifugal separator, aerators.
- KU3.** knowledge of code manual prescribed by BIS.
- KU4.** different types of meter and gauges.
- KU5.** working of different types of chemicals that are used during treatment of sewage.
- KU6.** knowledge of organic matter and inorganic matter.
- KU7.** different types of gas such as methane, carbon-di-oxide, hydrogen sulphide and many other gases.
- KU8.** knowledge of different types of graphs and readings.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or hindi/english
- GS2.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS4.**
 - read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or hindi
 -
- GS5.** state information, doubts and concerns about work related matters in local language or hindi/english
- GS6.** plan one's daily tasks to achieve maximum efficiency
- GS7.** be punctual and work as per agreed priorities
- GS8.** manage distractions and maintain workplace discipline
- GS9.** listen to the higher authorities carefully
- GS10.** be courteous
- GS11.** read and understand manuals, health and safety instructions, memos, other company documents.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>monitoring the stp</i>	15	30	-	5
PC1. able to monitor the ph, coagulation and flocculation, velocity of flow, discharge at inflow & outflow.	-	-	-	-
PC2. monitoring weather the biological process is being carried out as per the standards.	-	-	-	-
PC3. identify the mechanical instruments such as blowers, pumps, centrifugal separators, valves, aerators, etc are working properly	-	-	-	-
PC4. able to monitor the accumulation of sludge and pressure and temperature of digesters during evolution of gases while sludge digestion as per standards.	-	-	-	-
PC5. identify all the units in sewage treatment plant are in working condition.	-	-	-	-
PC6. able to monitor and record all meter and gauge readings.	-	-	-	-
<i>managing the stp</i>	15	30	-	5
PC7. able to identify that the screens are not clogged	-	-	-	-
PC8. identify any kind of leakage at joint of the screen and water channel.	-	-	-	-
PC9. able to examine the equalization tank that it should not overflow or leaking, mixing & aeration system.	-	-	-	-
PC10. able to identify the improper working of aeration system, mixing, leakage at any point, etc...,	-	-	-	-
PC11. able to prepare and maintain the record of operation carried out by sewage treatment plant.	-	-	-	-
PC12. able to fill out the daily activity log sheets.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. able to monitor the quality of water at the different units of the plants.	-	-	-	-
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0601
NOS Name	Monitoring and Managing Sewage Treatment Plant
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	4
Credits	3
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

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PSC/N0606: Solid Waste Management

Description

this unit is about the management of solid waste that is accumulated during the process of sewage treatment. It covers topics such as waste characterization, segregation techniques, recycling and reuse options, landfill management, and regulatory compliance. The lecture emphasizes the importance of proper waste management practices to minimize environmental impact, promote sustainability, and ensure compliance with waste management regulations.

Scope

The scope covers the following :

- prepare for disposal of waste
- disposal of waste

Elements and Performance Criteria

prepare for disposal for waste

To be competent, the user/individual on the job must be able to:

- PC1.** identification of probable sites for operating them as waste processing-cum-disposal sites.
- PC2.** able to identify the sites based on rapid eia
- PC3.** able to follow the criteria stipulated in the rules.
- PC4.** identify the method for the proper disposal of waste or sludge generated in plant according to the generation of waste.

disposal of solid waste

To be competent, the user/individual on the job must be able to:

- PC5.** able to compliance with regulatory requirements for solid waste disposal.
- PC6.** identify the ways of effective segregation of solid waste into different categories.
- PC7.** able to handle and store solid waste to prevent contamination and odors properly
- PC8.** identify efficient transportation of solid waste to disposal facilities.
- PC9.** able to minimize the solid waste generation through waste reduction and recycling initiatives.
- PC10.** able to implement the safe and secure storage areas for temporary waste storage.
- PC11.** able to carry out adequate monitoring and control of waste disposal operations.
- PC12.** able regular inspection and maintenance of disposal facilities to ensure functionality and safety.
- PC13.** able to implement the measures to prevent the release of hazardous substances from solid waste.
- PC14.** Able to perform the best practices for landfill management, including liner installation and leachate collection systems.
- PC15.** compliance with occupational health and safety guidelines during waste disposal activities.

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- PC16.** able to do regular evaluation and improvement of solid waste disposal processes and practices.
- PC17.** integration of sustainable waste management techniques, such as composting or anaerobic digestion.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** knowing the different types of solid waste such as domestic solid waste, industrial solid waste and biomedical solid waste.
- KU2.** knowledge of different types of domestic solid waste like garbage, rubbish, ashes, street sweeping.
- KU3.** understanding the different methods of municipal solid waste like sanitary landfilling method, gas production and its control, shredding & pulverization, barging/dumping into sea, composting, incineration & thermal pyrolysis, autoclaving.
- KU4.** knowledge of contamination and odors.
- KU5.** liner installation and leachate collection systems.
- KU6.** segregation of solid waste into different categories.
- KU7.** eia recommendations

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or hindi/english
- GS2.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** • perform arithmetic calculations of addition, subtraction, multiplication and division processes
•
- GS4.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or hindi
- GS5.** state information, doubts and concerns about work related matters in local language or hindi/english
- GS6.** plan one's daily tasks to achieve maximum efficiency
- GS7.** manage distractions and maintain workplace discipline
- GS8.** be punctual and work as per agreed priorities
- GS9.** listen to the higher authorities carefully
- GS10.** be courteous
- GS11.** read and understand manuals, health and safety instructions, memos, other company documents.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>prepare for disposal for waste</i>	10	20	-	3
PC1. identification of probable sites for operating them as waste processing-cum-disposal sites.	-	-	-	-
PC2. able to identify the sites based on rapid eia	-	-	-	-
PC3. able to follow the criteria stipulated in the rules.	-	-	-	-
PC4. identify the method for the proper disposal of waste or sludge generated in plant according to the generation of waste.	-	-	-	-
<i>disposal of solid waste</i>	20	40	-	7
PC5. able to compliance with regulatory requirements for solid waste disposal.	-	-	-	-
PC6. identify the ways of effective segregation of solid waste into different categories.	-	-	-	-
PC7. able to handle and store solid waste to prevent contamination and odors properly	-	-	-	-
PC8. identify efficient transportation of solid waste to disposal facilities.	-	-	-	-
PC9. able to minimize the solid waste generation through waste reduction and recycling initiatives.	-	-	-	-
PC10. able to implement the safe and secure storage areas for temporary waste storage.	-	-	-	-
PC11. able to carry out adequate monitoring and control of waste disposal operations.	-	-	-	-
PC12. able regular inspection and maintenance of disposal facilities to ensure functionality and safety.	-	-	-	-
PC13. able to implement the measures to prevent the release of hazardous substances from solid waste.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Able to perform the best practices for landfill management, including liner installation and leachate collection systems.	-	-	-	-
PC15. compliance with occupational health and safety guidelines during waste disposal activities.	-	-	-	-
PC16. able to do regular evaluation and improvement of solid waste disposal processes and practices.	-	-	-	-
PC17. integration of sustainable waste management techniques, such as composting or anaerobic digestion.	-	-	-	-
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0606
NOS Name	Solid Waste Management
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	4
Credits	4
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

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PSC/N0603: Maintain Records & Reporting at Site

Description

This unit focuses on the importance of maintaining accurate records and effectively reporting activities at a sewage treatment plant. It emphasizes the significance of documentation in ensuring compliance with regulations, monitoring plant performance, and facilitating continuous improvement. Trainees will learn how to record data, prepare reports, and communicate relevant information to stakeholders.

Scope

The scope covers the following :

- Understand how to maintain records and data
- Reporting to the higher officials
- Respond to higher officials

Elements and Performance Criteria

Understand how to maintain records and data

To be competent, the user/individual on the job must be able to:

- PC1.** able to read plan & specifications for different types of modifications in sewage treatment plant.
- PC2.** able to document data on pathogen reduction.
- PC3.** able to record the data on sludge accumulated and sludge treated.
- PC4.** able to maintain daily operational log.
- PC5.** perform routine operational duties, flow meter readings, sample information with corresponding analytical data, etc.
- PC6.** able to identify unusual conditions at the site
- PC7.** identify and report accident equipment hazard.
- PC8.** perform preventive maintenance

Reporting to the higher officials

To be competent, the user/individual on the job must be able to:

- PC9.** reporting to the supervisor if any kind of problem occurs
- PC10.** identify spare parts inventory when in need
- PC11.** identify specifications on equipment from supplier
- PC12.** equipment inventory

Respond to higher officials

To be competent, the user/individual on the job must be able to:

- PC13.** able to responding promptly to higher officials' inquiries or requests demonstrates a sense of urgency and professionalism.
- PC14.** able to provide clear and accurate information in responses to higher officials ensures effective communication and minimizes misunderstandings.

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- PC15.** able to ensure that responses align with relevant policies, procedures, and regulations demonstrates a commitment to adhering to organizational guidelines.
- PC16.** able to maintain respectful and courteous tone in responses conveys professionalism and enhances the organization's reputation.
- PC17.** identify potential solutions or recommendations in responses demonstrates proactive thinking and a commitment to finding effective resolutions.
- PC18.** able to follow up on requests or inquiries to higher officials and providing updates or additional information as needed helps to maintain effective communication and ensure that all concerns are addressed satisfactorily.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand local, state, and national regulations related to sewage treatment, including record-keeping and reporting requirements mandated by environmental agencies
- KU2.** Familiarity with different types of records, such as daily log sheets, maintenance logs, chemical usage reports, effluent quality records, energy consumption, and equipment performance logs.
- KU3.** Knowledge of how to sample influent and effluent water, record results from tests (e.g., pH, BOD, COD, TSS), and ensure these meet environment
- KU4.** Knowledge of keeping detailed logs of routine and emergency maintenance activities on equipment (e.g., pumps, aerators, clarifiers) to track equipment lifespan, repairs, and prevent failures.
- KU5.** Understand protocols for reporting spills, overflows, equipment malfunctions, or non-compliance events to regulatory authorities as required.
- KU6.** Knowledge of SOPs for different tasks within the sewage treatment plant, and the importance of maintaining updated records that align with these procedures.
- KU7.** Documenting the scheduled preventive maintenance tasks and keeping track of their completion to minimize downtime and optimize plant performance.
- KU8.** Knowledge to record data related to odor, noise, and emissions (if applicable) to ensure the plants environmental impact is minimized and within regulatory limits.
- KU9.** Understand the use of digital systems for data logging, report generation, and data retrieval. These systems may include SCADA (Supervisory Control and Data Acquisition) or other monitoring tools.
- KU10.** Generate performance reports for internal review, management oversight, and submission to regulatory bodies.
- KU11.** Knowledge of proper archiving of records, both in digital and hard copy formats, in accordance with legal retention policies, and ensuring easy retrieval during audits.
- KU12.** Knowledge of documentation procedures during emergencies (e.g., system failures, flooding) and having templates for swift reporting in urgent situations.

Generic Skills (GS)

User/individual on the job needs to know how to:



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- GS1.** prepare checklists, reports and fill out forms in local language or hindi/english
- GS2.** ability to read from different sources- books screens in machines and signage.
- GS3.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS4.** read and interpret information (symbols, dimensions, terminology, dates etc.) given in local language or hindi
- GS5.** state information, doubts and concerns about work related matters in local language or hindi/english
- GS6.** listen to the higher authorities carefully
- GS7.** communicate with supervisor.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understand how to maintain records and data</i>	10	20	-	3
PC1. able to read plan & specifications for different types of modifications in sewage treatment plant.	-	-	-	-
PC2. able to document data on pathogen reduction.	-	-	-	-
PC3. able to record the data on sludge accumulated and sludge treated.	-	-	-	-
PC4. able to maintain daily operational log.	-	-	-	-
PC5. perform routine operational duties, flow meter readings, sample information with corresponding analytical data, etc.	-	-	-	-
PC6. able to identify unusual conditions at the site	-	-	-	-
PC7. identify and report accident equipment hazard.	-	-	-	-
PC8. perform preventive maintenance	-	-	-	-
<i>Reporting to the higher officials</i>	10	20	-	4
PC9. reporting to the supervisor if any kind of problem occurs	-	-	-	-
PC10. identify spare parts inventory when in need	-	-	-	-
PC11. identify specifications on equipment from supplier	-	-	-	-
PC12. equipment inventory	-	-	-	-
<i>Respond to higher officials</i>	10	20	-	3
PC13. able to responding promptly to higher officials' inquiries or requests demonstrates a sense of urgency and professionalism.	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. able to provide clear and accurate information in responses to higher officials ensures effective communication and minimizes misunderstandings.	-	-	-	-
PC15. able to ensure that responses align with relevant policies, procedures, and regulations demonstrates a commitment to adhering to organizational guidelines.	-	-	-	-
PC16. able to maintain respectful and courteous tone in responses conveys professionalism and enhances the organization's reputation.	-	-	-	-
PC17. identify potential solutions or recommendations in responses demonstrates proactive thinking and a commitment to finding effective resolutions.	-	-	-	-
PC18. able to follow up on requests or inquiries to higher officials and providing updates or additional information as needed helps to maintain effective communication and ensure that all concerns are addressed satisfactorily.	-	-	-	-
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0603
NOS Name	Maintain Records & Reporting at Site
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	4
Credits	0.5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

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PSC/N0602: Health and Safety Practices at the treatment Plant

Description

this unit is about the health-related issues at stp and safety practices which workers need to be taken care off. health and safety practices plays a crucial role to ensure the well-being and protection of employees, visitors, and the environment. these practices aim to minimize the risks associated with handling and treating wastewater, hazardous chemicals, and potential biological contaminants.

Scope

The scope covers the following :

- Maintaining health at treatment plant
- safe work practises at the Treatment facilities

Elements and Performance Criteria

maintaining health at treatment plant

To be competent, the user/individual on the job must be able to:

- PC1.** identify the hazards associated with different tasks in a wastewater treatment plant.
- PC2.** able to understand and acquire know-how of safety measures to be taken when in project site.
- PC3.** identify general risks, including working at height and hazards associated with different tasks in a sewage treatment plant.
- PC4.** understand and acquire know-how of safety measures when handling mechanical parts in statutory or moving position.
- PC5.** perform the changing of chlorine cylinders safely.
- PC6.** perform the automatic and manual functions at scrubber unit safely as per instructions.

safe work practices at treatment facilities

To be competent, the user/individual on the job must be able to:

- PC7.** inform the higher authorities of the accident or hazard that has occurred.
- PC8.** select the right personal protective equipment for different tasks in a waste treatment plant.
- PC9.** ensure safe storage of corrosive substances.
- PC10.** follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with
- PC11.** able to clean and disinfect work area, materials/supplies, equipment etc.. before and after use.
- PC12.** able to identify hygiene and sanitation issues.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** possible causes of risk, hazard or accident in the workplace

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- KU2.** location of all the general health and safety equipment in the workplace
- KU3.** work practices and precautions to control and prevent risks, hazards and accidents
- KU4.** meaning of hazards and risks
- KU5.** do and donts in operation of waste treatment plant
- KU6.** practise of safe storage of corrosive substances.
- KU7.** able to detect chlorine leakage and control, scrubber unit gas leakage and control

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** prepare checklists, reports and fill out forms in local language or hindi/english
- GS2.** understand the various colour codes, as per standard electrical, mechanical and civil nomenclature.
- GS3.** perform arithmetic calculations of addition, subtraction, multiplication and division processes
- GS4.** state information, doubts and concerns about work related matters in local language or hindi/english
- GS5.** plan one's daily tasks to achieve maximum efficiency
- GS6.** be punctual and work as per agreed priorities
- GS7.** manage distractions and maintain workplace discipline
- GS8.** listen to the higher authorities carefully
- GS9.** be courteous
- GS10.** read and understand manuals, health and safety instructions, memos, other company documents

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>maintaining health at treatment plant</i>	15	30	-	5
PC1. identify the hazards associated with different tasks in a wastewater treatment plant.	-	-	-	-
PC2. able to understand and acquire know-how of safety measures to be taken when in project site.	-	-	-	-
PC3. identify general risks, including working at height and hazards associated with different tasks in a sewage treatment plant.	-	-	-	-
PC4. understand and acquire know-how of safety measures when handling mechanical parts in statutory or moving position.	-	-	-	-
PC5. perform the changing of chlorine cylinders safely.	-	-	-	-
PC6. perform the automatic and manual functions at scrubber unit safely as per instructions.	-	-	-	-
<i>safe work practices at treatment facilities</i>	15	30	-	5
PC7. inform the higher authorities of the accident or hazard that has occurred.	-	-	-	-
PC8. select the right personal protective equipment for different tasks in a waste treatment plant.	-	-	-	-
PC9. ensure safe storage of corrosive substances.	-	-	-	-
PC10. follow recommended personal hygiene and sanitation practices, for example, washing/sanitizing hands, covering face with	-	-	-	-
PC11. able to clean and disinfect work area, materials/supplies, equipment etc.. before and after use.	-	-	-	-
PC12. able to identify hygiene and sanitation issues.	-	-	-	-
NOS Total	30	60	-	10

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National Occupational Standards (NOS) Parameters

NOS Code	PSC/N0602
NOS Name	Health and Safety Practices at the treatment Plant
Sector	Plumbing
Sub-Sector	Water Supply & Water Treatment & Quality Control
Occupation	Plumbing Plant Operations
NSQF Level	4
Credits	0.5
Version	1.0
Last Reviewed Date	27/08/2024
Next Review Date	27/08/2027
NSQC Clearance Date	27/08/2024

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSC/N0604.Operating Sewage Treatment Plant (STP)	30	60	0	10	100	35
PSC/N0601.Monitoring and Managing Sewage Treatment Plant	30	60	0	10	100	15
PSC/N0606.Solid Waste Management	30	60	0	10	100	25
PSC/N0603.Maintain Records & Reporting at Site	30	60	0	10	100	10
PSC/N0602.Health and Safety Practices at the treatment Plant	30	60	0	10	100	10
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	5
Total	170	330	-	50	550	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
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National Occupational Standard	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by N
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding	Knowledge and Understanding (KU) are statements that together specify the technical, generic, professional and organisational specific knowledge that an individual need in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of those OS, these include communication related skills that are applicable to most job roles.
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Options	Options are NOS or set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.